END OF PHASE ONE PROEJECT

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INTRODUCTION

- Aviation Safety Risk Analysis
  Content:
- **Objective:** Analyze historical aviation incident data to determine whether, and how, our company should invest in aircraft — both commercial and private.
- **Scope:** Evaluate safety by aircraft make, flight purpose, and operational context.
- **Goal:** Support a data-driven market entry strategy into aviation.

BUSINESS CONTEXT

- **Key Question:** Should we enter the aviation sector at all?
- **Industry Insight:**
  - Aviation risk has dropped significantly over time.
  - Safe market niches and aircraft types exist — if selected carefully.
- **Opportunity:** Start with a low-risk, high-reliability segment (executive/business travel).

DATA SOURCE & STRUCTURE

- **Source:** National Transportation Safety Board (NTSB) accident database
- **Coverage:**
  - 1962–2023, global incidents in civil aviation
  - 90,000+ records; 30+ variables
- **Data Characteristics:**
  - Used: Injury severity, aircraft make, weather, flight purpose
  - Excluded: Variables with too many missing values
  - Focused analysis: Post-1980 data for reliability

PROCESS STEPS

- **Data Cleaning:** Filtered for relevant years, handled missing data
- **Normalization & Text Matching:** Standardized aircraft make (fuzzy matching)
- **Risk Index Creation:** Built using PCA from injury types
- **Exploratory Analysis:**
  - Time trends in safety
  - Risk heatmaps by purpose
  - Injury distribution by make
- **Visualization Tools:** Python, Matplotlib, Tableau

RESULTS & BUSINESS APPLICATION

- **Modern aviation is safer:** Injury rates down significantly since the 1980s
- **Low-risk entry point:**
  - Executive/business travel = lowest injury rate
  - Recommendation: phased market entry through this segment
- **Safe aircraft brands:**
  - Beech and Bellanca consistently low-risk among popular makes
  - Ideal candidates for initial procurement

EVALUATION & FUTURE IMPROVEMENTS

Limitations, Next Steps & Future Plans
Limitations:

  - Some data fields incomplete (e.g., weather, pilot characteristics)
  - Public dataset may not reflect newest models or private incidents

- **Next Steps:**
  - Validate insights with aviation experts
  - Engage procurement consultants for Beech/Bellanca availability
  - Build internal pilot protocols for travel routes & safety
  - Develop scoring tool to assess risk of future aircraft
  - Monitor incident trends and revise strategy accordingly